Numerical Methods for Elliptic Problems

Exercise Sheet 1

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Exercise 1: Vector identities and integration by parts

Let $\Omega \subset \mathbb{R}^d$ be bounded with piecewise smooth boundary, $\boldsymbol{b} \in C^1(\overline{\Omega}; \mathbb{R}^d)$ and $u, v \in C^2(\overline{\Omega})$. Write $\boldsymbol{n}$ for the outward unit normal and $\partial_n u = \nabla u \cdot \boldsymbol{n}$.

(a) By the product rule,

$$\begin{aligned}\operatorname{div}(u\boldsymbol{b}) &= \sum_{i=1}^{d} \partial_i(ub_i) = \sum_{i=1}^{d} ((\partial_i u) b_i + u \partial_i b_i) \\ &= \boldsymbol{b} \cdot \nabla u + u \operatorname{div} \boldsymbol{b}.\end{aligned}$$

(b) Applying the divergence theorem and then (a) gives

$$\int_{\partial\Omega} u\boldsymbol{b} \cdot \boldsymbol{n} \, \mathrm{d}s = \int_{\Omega} \operatorname{div}(u\boldsymbol{b}) \, \mathrm{d}x = \int_{\Omega} \boldsymbol{b} \cdot \nabla u \, \mathrm{d}x + \int_{\Omega} u \operatorname{div} \boldsymbol{b} \, \mathrm{d}x.$$

Rearranging yields

$$\int_{\Omega} \boldsymbol{b} \cdot \nabla u \, \mathrm{d}x = -\int_{\Omega} u \operatorname{div} \boldsymbol{b} \, \mathrm{d}x + \int_{\partial\Omega} u\boldsymbol{b} \cdot \boldsymbol{n} \, \mathrm{d}s.$$

(c) Choosing $\boldsymbol{b} = \nabla v$ in (b), we obtain

$$\int_{\Omega} \nabla u \cdot \nabla v \, \mathrm{d}x = -\int_{\Omega} u \Delta v \, \mathrm{d}x + \int_{\partial\Omega} u \partial_n v \, \mathrm{d}s.$$

Exchanging u and v and using symmetry of the inner product gives

$$\int_{\Omega} \nabla u \cdot \nabla v \, \mathrm{d}x = -\int_{\Omega} \Delta u \, v \, \mathrm{d}x + \int_{\partial\Omega} \partial_n u \, v \, \mathrm{d}s.$$

Exercise 2: Fundamental lemma, first version

Continuous case. Let $\Omega \subset \mathbb{R}^d$ be open and $f \in C(\overline{\Omega})$. If $f \equiv 0$, all the stated integrals vanish. Conversely, suppose $\int_\omega f \, \mathrm{d}x = 0$ for every bounded measurable $\omega \subset \Omega$. If $f(x_0) \neq 0$ at some $x_0 \in \Omega$, we may assume $f(x_0) > 0$ by replacing f by $-f$. By continuity and openness, there are $r > 0$ and $\varepsilon > 0$ such that $\overline{B}_r(x_0) \subset \Omega$ and $f(x) \geq \varepsilon$ on this ball. Hence

$$0 = \int_{\overline{B}_r(x_0)} f \, \mathrm{d}x \geq \varepsilon \, |\overline{B}_r(x_0)| > 0,$$

a contradiction. Thus $f \equiv 0$ in Ω.

L^1 case. If $f = 0$ almost everywhere, the integrals vanish. For the converse, let $f \in L^1(\Omega)$ satisfy the integral condition. For $n, R \in \mathbb{N}$ define the bounded measurable sets

$$E^+_{n,R} = \{x \in \Omega \cap B_R(0) : f(x) \geq 1/n\}, \qquad E^-_{n,R} = \{x \in \Omega \cap B_R(0) : f(x) \leq -1/n\}.$$

Then

$$0 = \int_{E^+_{n,R}} f \, \mathrm{d}x \geq \frac{1}{n} |E^+_{n,R}|, \qquad 0 = \int_{E^-_{n,R}} f \, \mathrm{d}x \leq -\frac{1}{n} |E^-_{n,R}|.$$

Consequently both sets have measure zero. Since

$$\{x \in \Omega : f(x) \neq 0\} = \bigcup_{n,R \in \mathbb{N}} (E^+_{n,R} \cup E^-_{n,R}),$$

countable subadditivity implies $f = 0$ almost everywhere in Ω.

Exercise 3: Fundamental lemma, second version

Continuous case. Let Ω be bounded and $f \in C(\overline{\Omega})$. The implication from $f \equiv 0$ to $\int_\Omega fv\,\mathrm{d}x = 0$ is immediate. Conversely, suppose the integral vanishes for all $v \in C(\overline{\Omega})$ with $v|_{\partial\Omega} = 0$. If $f(x_0) \neq 0$ for some $x_0 \in \Omega$, assume without loss of generality that $f(x_0) > 0$. Choose $r > 0$ such that $\overline{B}_r(x_0) \subset \Omega$ and $f > 0$ on this closed ball. Set

$$v(x) = \max\left\{1 - \frac{|x - x_0|}{r},\, 0\right\}.$$

This function is continuous, nonnegative, satisfies $v(x_0) = 1$, and vanishes outside $B_r(x_0)$, so it is an admissible test function. The product fv is strictly positive on $B_r(x_0)$ and zero outside. Therefore

$$\int_\Omega fv\,\mathrm{d}x = \int_{B_r(x_0)} fv\,\mathrm{d}x > 0,$$

contradicting the assumption. Thus $f(x) = 0$ for every $x \in \Omega$.

L^2 case. Let $f \in L^2(\Omega)$ and assume $\int_\Omega f\varphi\,\mathrm{d}x = 0$ for every $\varphi \in C_c^\infty(\Omega)$. For an arbitrary $v \in L^2(\Omega)$, density provides a sequence $\varphi_n \in C_c^\infty(\Omega)$ with $\|v - \varphi_n\|_{L^2(\Omega)} \to 0$. Cauchy–Schwarz gives

$$\left|\int_\Omega fv\,\mathrm{d}x\right| = \left|\int_\Omega f(v - \varphi_n)\,\mathrm{d}x\right| \leq \|f\|_{L^2(\Omega)}\|v - \varphi_n\|_{L^2(\Omega)} \longrightarrow 0.$$

Thus the integral vanishes for every $v \in L^2(\Omega)$. In particular, choosing $v = f$ yields

$$0 = \int_\Omega f^2\,\mathrm{d}x = \|f\|_{L^2(\Omega)}^2,$$

so $f = 0$ almost everywhere. Conversely, $f = 0$ almost everywhere implies the integral condition for all L^2 test functions, and hence for all C_c^∞ test functions as well.

Exercise 4: Euler–Lagrange equation

Let Ω be bounded with Lipschitz boundary, $a \in L^\infty(\Omega)$ with $a \geq a_0 > 0$ almost everywhere, and $f \in L^2(\Omega)$. We minimize

$$J(u) = \frac{1}{2}\int_\Omega a|\nabla u|^2\,\mathrm{d}x - \int_\Omega fu\,\mathrm{d}x \qquad \text{over } u \in H_0^1(\Omega).$$

(a) For $u, v \in H_0^1(\Omega)$, expansion of the square yields

$$J(u + tv) - J(u) = t\left(\int_\Omega a\nabla u \cdot \nabla v\,\mathrm{d}x - \int_\Omega fv\,\mathrm{d}x\right) + \frac{t^2}{2}\int_\Omega a|\nabla v|^2\,\mathrm{d}x.$$

Since $\left|\int_\Omega a|\nabla v|^2\,\mathrm{d}x\right| \leq \|a\|_{L^\infty(\Omega)}\|\nabla v\|_{L^2(\Omega)}^2 < \infty$, the first variation is

$$\delta J(u)[v] = \lim_{t\to 0}\frac{J(u + tv) - J(u)}{t} = \int_\Omega a\nabla u \cdot \nabla v\,\mathrm{d}x - \int_\Omega fv\,\mathrm{d}x.$$

At a minimizer it must vanish in every admissible direction. The variational problem is therefore: find $u \in H_0^1(\Omega)$ such that

$$\int_\Omega a\nabla u \cdot \nabla v\,\mathrm{d}x = \int_\Omega fv\,\mathrm{d}x \qquad \text{for every } v \in H_0^1(\Omega).$$

For comparison, the minimal surface problem has first-order condition

$$\int_\Omega \frac{\nabla u \cdot \nabla v}{\sqrt{1+|\nabla u|^2}}\,\mathrm{d}x = 0 \qquad \text{for every } v \in H_0^1(\Omega).$$

Here the left-hand side is bilinear in u and v, whereas the minimal surface equation is nonlinear in u. In addition, the present problem has the forcing term $\int_\Omega fv\,\mathrm{d}x$.

(b) Assume in addition $a \in C^1(\Omega)$, $f \in C(\Omega)$, and $u \in C^2(\Omega)$. Restricting to $v \in C_c^\infty(\Omega) \subset H_0^1(\Omega)$, integration by parts gives

$$\int_\Omega fv\,\mathrm{d}x = \int_\Omega a\nabla u \cdot \nabla v\,\mathrm{d}x = -\int_\Omega \operatorname{div}(a\nabla u)\,v\,\mathrm{d}x.$$

There is no boundary contribution because v has compact support in Ω. Hence

$$\int_\Omega (f + \operatorname{div}(a\nabla u))v\,\mathrm{d}x = 0 \qquad \text{for every } v \in C_c^\infty(\Omega).$$

The fundamental lemma implies

$$-\operatorname{div}(a\nabla u) = f \quad \text{in } \Omega, \qquad u = 0 \quad \text{on } \partial\Omega.$$

The boundary condition follows from the admissible space, in the trace sense (and pointwise whenever u extends continuously to the boundary).

Exercise 5: Homogenizing Dirichlet data

Let $g = u_g|_{\partial\Omega}$ for $u_g \in C^2(\overline{\Omega})$, and let $u \in C^2(\Omega) \cap C(\overline{\Omega})$ solve

$$-\Delta u = 0 \quad \text{in } \Omega, \qquad u = g \quad \text{on } \partial\Omega.$$

Choose $w = u - u_g$. Then

$$-\Delta w = -\Delta u + \Delta u_g = \Delta u_g \quad \text{in } \Omega,$$

and $w|_{\partial\Omega} = g - g = 0$. Thus $u = u_g + w$, with

$$-\Delta w = f \quad \text{in } \Omega, \qquad w = 0 \quad \text{on } \partial\Omega, \qquad f = \Delta u_g.$$

The right-hand side depends only on the chosen extension u_g.

Exercise 7: Local NGSolve installation

The local NGSolve installation and C++ extension test were completed. The recorded notebook run compiled `mymodule.cpp` using `CompilePythonModule` and passed the following checks:

```
assert m.answer() == 42
mesh = Mesh(unit_square.GenerateMesh(maxh=0.2))
fes = H1(mesh, order=2)
assert m.ndof(fes) == fes.ndof
```

The final output was `C++ extension test passed.`